1. Clinical Question & Core Hypothesis

Clinical Question

From the Ontologist's synthesis, TMB and local immune contexture (CD4/CD8 TILs) are each linked to outcome in MIBC, yet their prognostic/predictive value is inconsistent and implementation is constrained by accessibility; the added observation of tumor-derived SDC2/TFPI2 methylation in stool suggests an epigenetic/ECM phenotype that could modulate immune-tumor interactions. 2) Tier 1 paths used: (a) TMB — chosen because it mechanistically modifies neoantigen load and is the best molecular determinant of ICI sensitivity; (b) SDC2/TFPI2 methylation — chosen as a tractable, non-invasive epigenetic marker that signals an altered ECM/tumor-shedding phenotype potentially linked to immune exclusion; (c) tissue TILs (operationalized here as image-derived TIL density) — chosen because it is the direct morphologic readout of local adaptive immunity. I built the hypothesis around how TMB and SDC2-related methylation modify the functional impact of image-measured TIL density. 3) Tier 2 support: AI-enabled biomarker accessibility and MSI strengthen the translational path (making combined TIL/TMB/methylation testing feasible); HDL-C and cross-cancer biomarkers (PSA) provide orthogonal metabolic and implementation context but not mechanistic primacy. 4) Why this endpoint: I selected therapy_response (immune checkpoint blockade response proxy) because TMB and TIL density are mechanistically tied to anti-tumor immune recognition and hence to ICI efficacy — this yields the most testable, clinically actionable question. 5) Deliberate exclusions: I excluded using PSA, pZVR, CSF-AD links and other cross-disease signals as core anchors because they lack direct mechanistic connection to MIBC immune biology; TFPI2, CD4, CD8, MSI, HDL-C and AI-access methods are retained as secondary_modifiers for follow-up stratified analyses rather than primary anchor variables to preserve focus?

Working Hypothesis

In muscle-invasive bladder cancer, the predictive value of high tumor-infiltrating lymphocyte (TIL) density for immune checkpoint inhibitor (ICI) response is conditional on tumor mutational burden (TMB); high TMB amplifies TIL-mediated anti-tumor activity, whereas SDC2 promoter methylation marks an ECM-remodeling/immune-exclusion phenotype that impairs TIL effector function. Consequently, a composite biomarker integrating tissue TIL density, tissue TMB, and SDC2 methylation status (and optionally stool SDC2/TFPI2 methylation as a non-invasive readout) will better stratify ICI responders. Specifically, the subset with TIL-high, TMB-high, and SDC2 unmethylated tumors will exhibit the highest likelihood of objective response, while TIL-high with TMB-low or SDC2-methylated status will show attenuated or absent benefit despite abundant TILs.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	Tumour Mutational Burden (TMB), SDC2
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

Tumour Mutational Burden (TMB): Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

SDC2: SDC2/TFPI2 methylation — chosen as a tractable, non-invasive epigenetic marker that signals an altered ECM/tumor-shedding phenotype potentially linked to immune exclusion; (c) tissue TILs (operationalized here as image-derived TIL density) — chosen because it is the direct morphologic readout of local adaptive immunity.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	TFPI2	Supporting modifier from knowledge graph
Secondary	CD4	Supporting modifier from knowledge graph
Secondary	CD8	Supporting modifier from knowledge graph
Secondary	MSI	Supporting modifier from knowledge graph
Secondary	AI-based biomarker accessibility	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: Retrospective cohort analyses with quantified TIL density by digital pathology, tumor TMB (via whole-exome or targeted panels), and tumor SDC2 promoter methylation;

Prediction 2: stool assays for SDC2/TFPI2 methylation as non-invasive proxies;

Prediction 3: statistical modeling to test interaction terms: TIL density × TMB, and both × SDC2 methylation, plus development of a composite score;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TFPI2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CD4: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.